

Felicity Lester

1.a

	A	B	C	D
P1	3.5 (1)	3.5 (2)	3.5 (3)	3.5 (2)
P2	2.5 (3)	2.5 (2)	2.5 (1)	2.5 (1)
	15%	20%	55%	10%
	0.15	0.20	0.55	0.10

$$\text{GHz} \rightarrow \text{Hz} = 10^9$$

$$\text{Effective CPI} = 1 \times 0.15 + 2 \times 0.20 + 3 \times 0.55 + 2 \times 0.10 = \boxed{2.4}$$

$$\text{P2} = 3 \times 0.15 + 2 \times 0.2 + 1 \times 0.55 + 1 \times 0.10 = \boxed{1.5}$$

$$\text{1.b CPU Time P1} = (2 \times 10^6) \times 2.4 \times \frac{1}{3.5 \times 10^9} \approx .0013714 = \boxed{1.3714 \text{ ms}}$$

$$\text{CPU Time P2} = (2 \times 10^6) \times 1.5 \times \frac{1}{2.5 \times 10^9} = .0012 = \boxed{1.2 \text{ ms}}$$

P2 is faster

1.c IC x Effective CPI

$$\text{P1} = (2 \times 10^6) \times 2.4 = \boxed{4.8 \times 10^6}$$

$$\text{P2} = (2 \times 10^6) \times 1.5 = \boxed{3.0 \times 10^6}$$

$$1.d \quad P1 = \frac{(2 \times 10^6)}{1.3714} = \boxed{1.459 \times 10^9}$$

$$P2 = \frac{(2 \times 10^6)}{1.2} = \boxed{1.667 \times 10^9}$$

$$1.e \quad P1 = \frac{(1.25) \times 2.4}{0.7} = 4.2856 \times 10^9$$

$$P2 = \frac{(1.25) \times 1.5}{0.7} \times 2.5 = 6.6964 \times 10^9$$

reduce by 30% = 0.7

increase by 25% = 1 + 0.25 = 1.25

Q2

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$$X \rightarrow IC_x = 8 \times 10^8 -$$

$$ET_x = 1.2 \text{ sec}$$

$$Y \rightarrow IC_y = 1.6 \times 10^9 -$$

$$ET_y = 2.4 \text{ sec}$$

$$\frac{1}{3 \times 10^9} \dots 3.33 \times 10^8$$

$$2.a \quad XCP1 \Rightarrow \frac{1.2 (3 \times 10^9) (3.33 \times 10^8)}{8 \times 10^8} = \boxed{0.4995}$$

$$YCP1 \Rightarrow \frac{2.4 (3.33 \times 10^8)}{1.6 \times 10^9} = \boxed{0.4995}$$

$$2.b \quad \frac{8 \times 10^8 \times 0.4995}{1.6 \times 10^9 \times 0.4995} = \boxed{0.5}$$

X is 0.5 (half) of Y

$$\text{Speed } V_P = \frac{\text{old}}{\text{New}}$$

$$2.c \quad \frac{2.5^{\text{NEW}} \times 1.3 \times 10^9}{3.33 \times 10^8} = \boxed{9.76} \text{ slowest}$$

$$X \quad \frac{1.2}{9.76} = \boxed{0.12295} \text{ fastest}$$

$$Y \quad \frac{2.4}{9.76} = \boxed{0.2459}$$

Q3

Clock Rate = 3.2 GHz

Voltage = 1.0 V

35 W static (SP)

45 W dynamic (DP)

$$DP = \frac{3.2 \times 10^9 \times 1^2 \times}{2}$$

3.a

$$\text{Capacitive Load} = \frac{45}{(1.0)^2 \times 3.2 \times 10^9} = 1.40625 \times 10^{-8}$$

$$\downarrow$$

$$\boxed{14.06 \text{ fF}}$$

$$3.b \quad TP = 35 + 45 = 80$$

$$\frac{35}{45} \times 100\% = \boxed{43.75\% \text{ Total Power}}$$

$$\frac{SP}{DP} = \frac{35}{45} = \boxed{.7778 \text{ Ratio}}$$

$$3.c \quad TP_{\text{New}} = 80 \times (1 - 0.15) = 68 \text{ W}$$

$$DP_{\text{New}} = 68 - 35 = 33 \text{ W}$$

$$\frac{33}{(1.40625 \times 10^{-8}) \times (3.2 \times 10^9)} \approx 0.733$$

$$V_{\text{New}} = \sqrt{0.73333} \approx \boxed{0.856 V_{\text{New}}} \text{ reduced}$$

Q4 IC 4.0×10^{12}

ET 700s

RT 9000s

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$$\frac{9000}{700} = \boxed{12.857 \text{ SPEC ratio}}$$

4.a clock cycle $\rightarrow 0.5 \text{ ns} = 2 \text{ GHz}$

$$\text{CPI} = \frac{700 \times (2 \times 10^9)}{4.0 \times 10^{12}} = \boxed{0.35 \text{ CPI}}$$

4.b. 0.25% increase

$$(1 + 0.25) \times 4.0 \times 10^{12} = 5 \times 10^{12}$$

$$\text{New CPU Time} = \frac{5 \times 10^{12} \times 0.35}{2 \times 10^9} = 875 \text{ sec.}$$

$$\text{New SPEC ratio} = \frac{\text{RT}}{\text{New T}} = \frac{9000}{875} = \boxed{10.29_{\text{New}}}$$

4.c Instruction $\uparrow$ 10%
CPI $\uparrow$ 15%

$$(1+0.10) \times 4.0 \times 10^{12} = 4.4 \times 10^{12}$$

$$CPI_{\text{new}} = (1+0.15) \times (0.35) = 0.4025$$

$$\text{CPU Time} = \frac{4.4 \times 10^{12} \times 0.4025}{2 \times 10^9} = 885.5 \text{ sec.}$$

$$\text{New SPECratio} = \frac{RT}{\text{New ET}} = \frac{9000}{885.5} \approx \boxed{10.164}$$

4.d 4GHz = 4×10^9

$$IC - 0.10\% \quad (1-0.10) \times 4 \times 10^{12} = 3.6 \times 10^{12}$$

ET - 650s

SPECratio - 13.85

$$\text{New CPI} = \frac{650s \times (4 \times 10^9)}{3.6 \times 10^{12}} = \boxed{0.7222}$$

Original CPU = 700 +
New is 650

$$\frac{700-650}{700} = .0714 \times 100$$

= $\boxed{7.14\% \text{ CPU reduction}}$

4.a = 0.35CPI $\rightarrow$ in 4.d CPI increased to 0.7222

Original CR = 2GHz $\rightarrow$ New = 4GHz so 100% increase
is similar to CPI which has a 106% increase

Q5

Time 300s

FP 70s

INT 100s

LIS 65s

Branching 65s

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$$5.a \quad 0.25 \times 70s (FP) = \boxed{17.5 \text{ sec. New Time}}$$

$$70 - 17.5 = 52.5 \text{ sec}$$

$$\text{New Time} = 300 - 17.5 = 282.5 \text{ sec}$$

$$\text{Reduction} = \frac{300 - 282.5}{300} = \boxed{5.833 \text{ reduction}}$$

$$5.b \quad 300 \times (1 - 0.15)^{15\%} = 255s$$

$$300 - 255 = 45s$$

$$\text{INT} = 100 - 45s = \boxed{55 \text{ sec reduction}}$$

$$5.c \quad 300 \times (1 - 0.25)^{25\%} = 225s$$

$$300 - 225 = 75s$$

$$0.25 \times 65 = 16.25 \text{ BT reduction}$$

$$65 - 16.25 = 48.75s \text{ New BT}$$

$$300 - 16.25 = 283.75s \text{ New Time}$$

$$\frac{300 - 283.75}{300} \times 100 = \boxed{5.4166\% \text{ less than original}} \\ \text{NOT } 25\%$$